



Vivekanand Education Society's Institute of Technology

An Autonomous Institute Affiliated to University of Mumbai

End Semester Examination Special Exam – 2024 - 2025

Max marks: 60

Branch: Computer Engineering

Name of the Course: Artificial Intelligence

Course code: NCMMM41

Duration: 2 hours

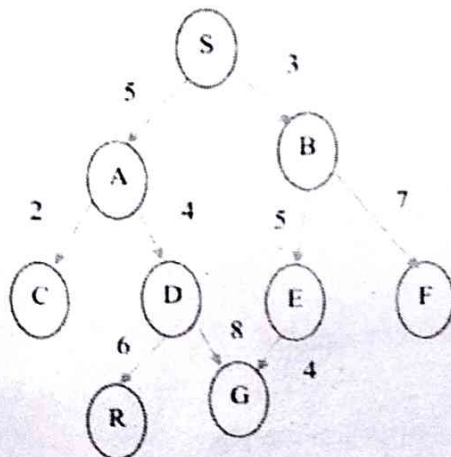
Semester: IV

QP Code: R24-NCMMM41_022024-25

- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary.

Marks

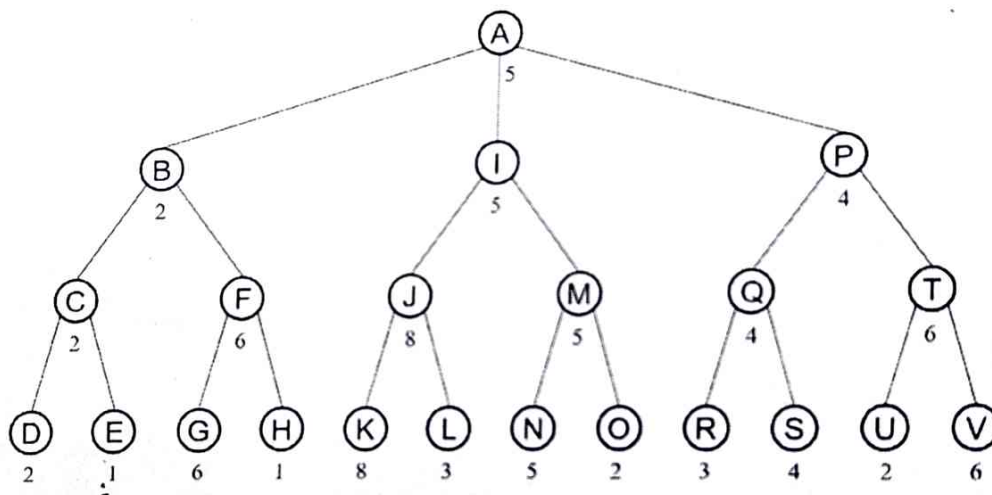
- (a) Differentiate between a simple reflex agent and a goal-based agent. 05
- (b) Consider a medical diagnosis system as an AI agent. Formulate its PEAS descriptor and explain its components. 05
- (c) Apply UCS to the following graph. Find the optimal path from node S to node G. 05
Estimate the path cost



- (d) Translate the following statements into propositional logic with an explanation: 05
1. "If it rains, the ground gets wet."
 2. "If the ground is not wet, then it did not rain."
 3. "The ground is wet only if it rains."

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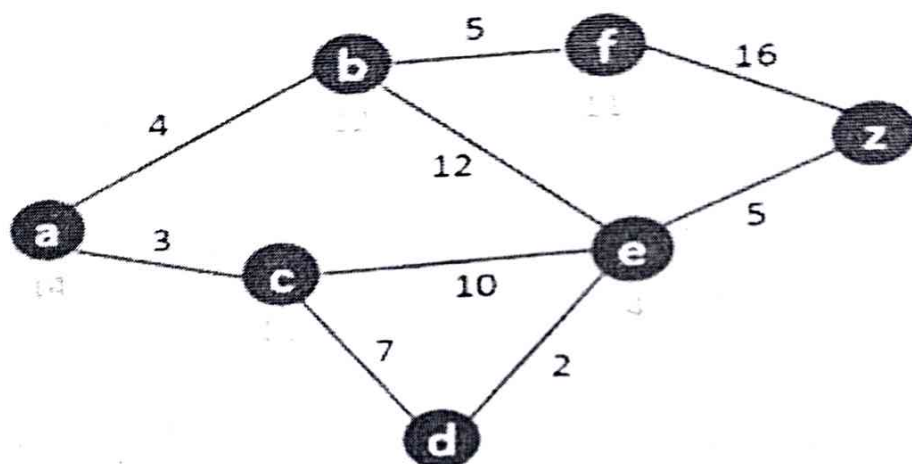
- Q.2 (a) Explain the fundamental concept of a genetic algorithm. Describe the main components of a genetic algorithm using a flowchart. Give two applications of it.
- (b) For the given game tree, apply the alpha-beta pruning algorithm starting from node A. Traverse the complete tree and show the pruned branches clearly. Identify the optimal value at the root node and write the optimal path. How it is advantageous compared to the MiniMax algorithm?



- Q.3 (a) How is FOL better than propositional logic? Write FOL statements to express the following conditions:
1. Every doctor works in a hospital.
 2. There exists a doctor who works in two different hospitals.
 3. No doctor works in all hospitals.



- (b) Apply the A* search to find the solution path from node 'a' to 'z'. Write all steps as well as open and closed lists. Determine the optimal path and root node value. 10



- (a) Differentiate environments: 10
- Fully observable vs Partially observable
 - Single agent vs Multiagent
 - Deterministic vs Stochastic
 - Static vs Dynamic
- (b) Describe the forward chaining and backward chaining inference methods in FOL. 10
Provide examples for both.
- (a) Discuss the structure and working of a Bayesian Belief Network with a suitable example. 10
- (b) Explain supervised, unsupervised, semi-supervised, and reinforcement learning with suitable applications. 10

X—X—X—X